

Project Title
Delivery Route Optimizer

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The **Delivery Route Optimizer** is a Java-based project developed to improve the efficiency of delivery operations by identifying the shortest and most cost-effective routes between multiple locations. The system applies Data Structures and Algorithms (DSA) concepts, particularly graph representations and shortest path algorithms such as Dijkstra's Algorithm, to calculate the optimal delivery path. Users can enter delivery locations, distances, and destination details, and the application processes this information to generate the best possible route while minimizing travel distance and delivery time. The project demonstrates the practical application of graphs, priority queues, and efficient searching techniques in solving real-world logistics problems. It also provides a simple and user-friendly interface for entering route information and displaying optimized results. By reducing unnecessary travel and improving route planning, the system helps save time, fuel, and operational costs. This project enhances understanding of algorithm design, graph data structures, and optimization techniques while showcasing how DSA can be applied to modern transportation and logistics challenges. The Delivery Route Optimizer serves as an educational and practical solution for efficient route management.

Signature of the Guide